

# Microeconomics

## Lecture 15: Taxes and Subsidies

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# What We Know

## Tools for welfare analysis:

- CS, PS, TS = CS + PS
- First Welfare Theorem: the competitive equilibrium maximizes TS
- Deadweight loss (DWL) as a measure of the inefficiency of an intervention
- Triangle formula:  $DWL = \frac{1}{2}(Q^* - \tilde{Q})(P_D(\tilde{Q}) - P_S(\tilde{Q}))$

**Administered prices:** both the *ceiling* and the *floor* distort the quantity traded and generate DWL, but they generate no revenue for the government. Today we study interventions that **produce monetary transfers**.

## In this lecture

How do **taxes** and **subsidies** affect equilibrium and welfare? Two key questions:

1. Who *really* pays a tax? (The seller who remits it or the buyer?)
2. Why does even a subsidy—which looks like a “gift”—produce a deadweight loss?

## Lecture Objectives

1. **Distinguish** specific, ad valorem, and lump-sum taxes
2. **Explain** the shifting mechanism of a specific tax
3. **Compute** the new equilibrium, CS, PS, revenue, and DWL after a tax
4. **Determine** tax incidence and relate it to elasticities
5. **Prove** the Equivalence Principle: incidence does not depend on who remits the tax
6. **Analyze** the relationship between DWL and the tax rate ( $DWL \propto t^2$ )
7. **Study** a subsidy and its symmetry with a tax
8. **Solve** complete numerical exercises

### Reference market

$Q^D = 80 - 2P$  (inv.  $P_D = 40 - Q/2$ ) and  $Q^S = 3P$  (inv.  $P_S = Q/3$ ).

Equilibrium:  $P^* = 16$ ,  $Q^* = 48$ , CS = 576, PS = 384, TS = 960.

# Why Tax? Three Functions of Taxes

Taxes perform three analytically distinct functions, often intertwined in practice:

1. **Allocative function.** They finance the production of public goods and address market failures (Pigouvian taxes on negative externalities). Here the tax *corrects* an inefficiency.
2. **Redistributive function.** They transfer resources across taxpayers (progressive IRPEF, inheritance taxes). Here the tax accepts inefficiency as the price of equity.
3. **Pure financing function.** They raise revenue to pay for public services (education, health care, defense). Here the tax distorts the market without correcting anything.

**Our focus in this lecture.** We analyze the *microeconomic* effects of taxes—**how** they distort equilibrium, **how much** they cost in terms of DWL, **who** really bears them. Choosing among the three functions is a *welfare economics* question beyond this course.

## General rule

All taxes (except lump-sum ones) generate DWL. The relevant question is not “how to avoid DWL”, but “how to minimize it for a given revenue”. This is the *optimal taxation* problem, a classic of public economics.

# Three Types of Tax

## 1. Specific tax (per unit, *excise tax*)

A fixed amount  $t$  for each unit produced or sold.

**Examples:** excise duty on gasoline (/liter), tobacco tax (/pack).

## 2. Ad valorem tax (percentage)

A rate  $\tau$  applied to the price: the amount levied varies with the price.

**Examples:** VAT at 22%, percentage customs duties.

## 3. Lump-sum tax (fixed sum)

A fixed amount independent of quantity and price.

**Examples:** license fee, flat-rate wealth tax.

## Key difference for efficiency

A **lump-sum** tax does not change  $MC$  and hence does not alter the choice of  $Q$ :  $DWL = 0$ . **Specific** and **ad valorem** taxes change  $MC$ , hence the supply curve and  $Q^*$ :  $DWL > 0$ .

## Lump-Sum Tax: No Distortion

**Why does a lump-sum tax not distort?** Suppose a firm pays a fixed tax  $F$  regardless of the quantity produced.

- At  $Q = 0$ : it pays  $F$
- At  $Q = 10$ : it pays  $F$
- At  $Q = 100$ : it pays  $F$

The marginal cost of producing one more unit does not change: the condition  $P = MC$  determines the same  $Q^*$  as before. The firm chooses the same output level; only profit changes ( $\pi - F$ ).

### The British Poll Tax (1989–1991)

Margaret Thatcher introduced a fixed tax per adult (*Community Charge*). Economically efficient ( $DWL = 0$ ), but politically unsustainable: unfair and regressive (a poor citizen pays the same amount as a rich one). Repealed in 1991 after violent protests. *Pareto-technical* efficiency is not the only criterion for evaluating public policies.

**In practice:** governments rarely use lump-sum taxes precisely because of the social-acceptability constraint. The efficiency–equity trade-off comes back in almost every *public finance* choice.

## Specific Tax: Mechanism

**Specific tax  $t$  on the producer.** For each unit sold at price  $P_d$  (paid by the buyer), the producer remits  $t$  to the government and keeps  $P_s = P_d - t$ .

**Effect on the supply curve:** the producer supplies  $Q$  only if the after-tax price covers  $MC$ :

$$P_d - t \geq MC(Q) \iff P_d \geq MC(Q) + t$$

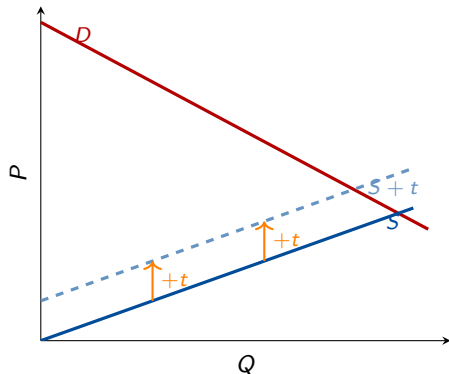
The supply curve shifts **vertically upward by  $t$** : to produce the same  $Q$ , producers now ask for  $t$  more.

With  $t = 5$  in our market

New supply:  $P_s = P_d - 5 \Rightarrow Q_t^S = 3(P_d - 5) = 3P_d - 15$ .

**Notation.** We use  $P_d$  for the price paid by the buyer and  $P_s$  for the price kept by the producer. Without a tax  $P_d = P_s = P^*$ . With a tax  $P_d - P_s = t$  (the “tax wedge”).

## Specific Tax: The Supply Shift



Every point of curve  $S$  shifts upward by  $t$ , producing  $S + t$ .

**Interpretation:** at any quantity  $Q$ , the *new* price asked by the seller is the old price  $P_S(Q)$  plus the tax  $t$ .

**The new equilibrium** is at the intersection of demand  $D$  (unchanged, it depends on the price paid  $P_d$ ) and the new supply  $S + t$ : the point where  $D = S + t$ .

**From that intersection:**

- $P_d^*$  (vertical coordinate) = price paid by the consumer
- $Q_t^*$  (horizontal coordinate) = quantity traded
- $P_s^* = P_d^* - t$  = price kept by the producer (read off the original curve  $S$ )

## Specific Tax: New Equilibrium

**Equilibrium with a tax  $t = 5$  on the producer:**

$$Q^D = Q_t^S \iff 80 - 2P_d = 3P_d - 15 \iff 95 = 5P_d$$

$$P_d^* = 19, \quad P_s^* = 19 - 5 = 14, \quad Q_t^* = 80 - 38 = 42$$

**Check:**  $Q^D(19) = 42$ ;  $Q_t^S = 3 \cdot 14 = 42 \checkmark$

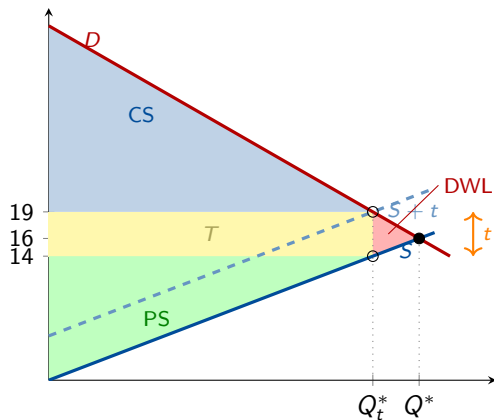
**Economic reading:**

- Buyers pay  $P_d^* = 19$  (was 16): +3 per unit
- Producers receive  $P_s^* = 14$  (was 16): -2 per unit
- Quantity falls from 48 to 42
- The tax wedge  $t = 5$  splits into  $3 + 2$
- Tax revenue:  $T = t \cdot Q_t^* = 5 \times 42 = 210$

### First incidence surprise

The tax is legally levied on the *producer*, but *consumers* bear the larger share (3/5). Whoever formally remits the tax is not necessarily whoever bears it economically.

## Tax $t = 5$ : Graph with All the Areas



**Components at  $t = 5$ ,  $Q_t^* = 42$ :**

$$CS = \frac{1}{2} \cdot 42 \cdot (40 - 19) = 441$$

$$PS = \frac{1}{2} \cdot 42 \cdot 14 = 294$$

$$T = t \cdot Q_t^* = 5 \cdot 42 = 210$$

$$DWL = \frac{1}{2} \cdot (48 - 42) \cdot 5 = 15$$

**Overall check:**

$$CS + PS + T + DWL = 960 \checkmark$$

**Changes:**

$$\Delta CS = -135, \Delta PS = -90$$

$$\text{Revenue } T = +210, DWL = +15$$

Algebraic sum:

$-135 - 90 + 210 = -15 = -DWL$ : cons. and prod. lose 225; the government collects 210; residual 15 = DWL.

## Tax: Step-by-Step Decomposition

Starting from the equilibrium areas, we see how each one is transformed by the introduction of the tax.

| Area                 | Transformation with $t$                   | Before | After      |
|----------------------|-------------------------------------------|--------|------------|
| CS                   | Smaller triangle above $P_d^*$            | 576    | 441 (−135) |
| PS                   | Smaller triangle below $P_s^*$            | 384    | 294 (−90)  |
| Tax wedge ( $T$ )    | New rectangle between $P_s^*$ and $P_d^*$ | 0      | +210       |
| DWL                  | New triangle to the right of $Q_t^*$      | 0      | +15        |
| Net TS (CS+PS+ $T$ ) |                                           | 960    | 945        |

### Area-by-area interpretation:

- CS loses 135: part becomes revenue (126, upper rectangle handed to the tax authority), part turns into DWL (9)
- PS loses 90: part revenue (84), part DWL (6)
- The **revenue rectangle** ( $T = 210$ ) is split exactly between the part taken from CS (126) and the part taken from PS (84). It is a pure transfer, not a loss.
- Only the **DWL triangle** (15) is value that vanishes: none of the three parties collects it.

# Who Really Pays the Tax?

## Tax incidence

**Tax incidence** describes how the economic burden of the tax is distributed between consumers and producers, regardless of who formally remits it to the government.

**In our example ( $t = 5$ ):**

- $\Delta P_d = 19 - 16 = +3$  (how much more consumers pay per unit)
- $|\Delta P_s| = |14 - 16| = 2$  (how much less producers receive per unit)
- $\Delta P_d + |\Delta P_s| = 3 + 2 = 5 = t \checkmark$

## Incidence shares

$$\text{Consumer share} = \frac{\Delta P_d}{t} = \frac{3}{5} = 60\%$$

$$\text{Producer share} = \frac{|\Delta P_s|}{t} = \frac{2}{5} = 40\%$$

# The Equivalence Principle

**What happens if, instead of taxing the seller, we tax the buyer?** Let us check with a tax on the buyer.

**With a tax  $t = 5$  on the buyer:** the buyer pays  $P_d$  to the seller and  $t$  to the government. Bearing  $P_d + t$  in total, the buyer will purchase only if his WTP justifies it. So the seller receives  $P_s = P_d$ , and demand becomes:

$$Q_t^D = 80 - 2(P_s + t) = 80 - 2P_s - 10 = 70 - 2P_s$$

**New equilibrium:**

$$70 - 2P_s = 3P_s \iff P_s^* = 14 \iff P_d^* = 14 + 5 = 19, Q_t^* = 42$$

## Equivalence Principle

The final equilibrium ( $P_d^* = 19$ ,  $P_s^* = 14$ ,  $Q_t^* = 42$ ) is **identical** to the case of the tax on the seller. CS, PS, revenue, and DWL are the same.

**Intuition.** The market automatically “shifts” the tax along the slopes of the D and S curves. The distinction between who formally remits and who economically bears the tax is purely an accounting one.

# Incidence and Elasticity: The General Rule

**General rule:** the **less elastic** side bears the larger share.

## Incidence formula

$$\text{Consumer share} = \frac{\varepsilon_S}{\varepsilon_S - \varepsilon_D}, \quad \text{Producer share} = \frac{-\varepsilon_D}{\varepsilon_S - \varepsilon_D}$$

(Recall  $\varepsilon_D < 0$  and  $\varepsilon_S > 0$ ; the signs ensure positive shares.)

**Check on our market** ( $P^* = 16$ ,  $Q^* = 48$ ):

$$\varepsilon_D = \frac{dQ^D}{dP} \cdot \frac{P}{Q} = -2 \cdot \frac{16}{48} = -\frac{2}{3}$$

$$\varepsilon_S = \frac{dQ^S}{dP} \cdot \frac{P}{Q} = 3 \cdot \frac{16}{48} = 1$$

$$\text{Cons. share} = \frac{1}{1 - (-2/3)} = \frac{1}{5/3} = \frac{3}{5} = 60\% \quad \checkmark$$

## Intuition

With *inelastic demand*, consumers do not cut quantities much as the price rises  $\Rightarrow$  they absorb most of the tax.  
With *inelastic supply*, producers do not cut output much  $\Rightarrow$  they absorb most of it.

## Intuitive Derivation of Incidence (1/2)

Why is the consumer share exactly  $\varepsilon_S/(\varepsilon_S - \varepsilon_D)$ ? Here is the idea *without implicit derivatives*.

**Step 1.** The tax  $t$  creates a “wedge” between prices:  $P_d - P_s = t$ . Let  $\Delta P_d$  denote how much the price paid by buyers rises and  $|\Delta P_s|$  how much the producers’ price falls. By construction  $\Delta P_d + |\Delta P_s| = t$ .

**Step 2.** For a given change in  $Q$ , the two sides make different “moves” along their curves:

- If  $D$  is **steep** (inelastic), a large  $\Delta P_d$  is needed to reduce demand by the given  $\Delta Q$
- If  $S$  is **flat** (elastic), a small  $|\Delta P_s|$  is enough to reduce supply by the same  $\Delta Q$

**Geometric intuition.** The side with the steeper curve is “stuck” moving along an almost vertical segment: its price change is large for a small change in quantity. This side bears the larger share of the tax.

## Intuitive Derivation of Incidence (2/2)

**Step 3.** Using the definition of elasticity, for small changes:

$$\Delta P_d \approx \frac{\Delta Q/Q^*}{\varepsilon_D} \cdot P^*, \quad \Delta P_s \approx \frac{\Delta Q/Q^*}{\varepsilon_S} \cdot P^*$$

(The consumer has  $\varepsilon_D < 0$ , hence  $\Delta P_d > 0$ ; the producer has  $\varepsilon_S > 0$  and  $\Delta Q < 0$ , hence  $\Delta P_s < 0$ .)

**Step 4.** Taking the ratio of the two changes in absolute value:

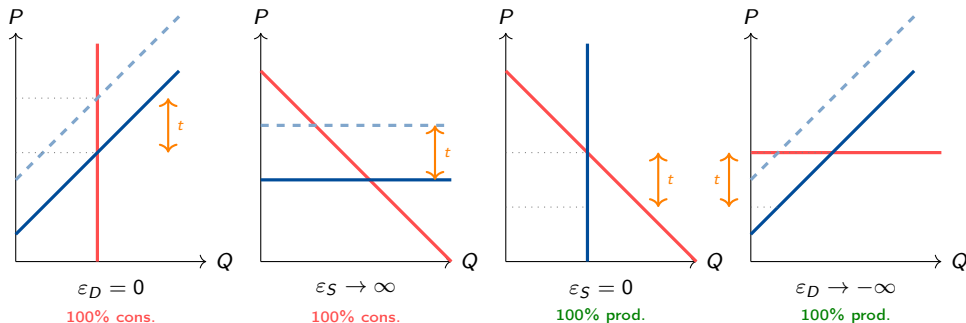
$$\frac{\Delta P_d}{|\Delta P_s|} = \frac{\varepsilon_S}{|\varepsilon_D|}$$

**Step 5.** From  $\Delta P_d + |\Delta P_s| = t$  and the ratio above:

$$\Delta P_d = \frac{\varepsilon_S}{\varepsilon_S + |\varepsilon_D|} \cdot t = \frac{\varepsilon_S}{\varepsilon_S - \varepsilon_D} \cdot t \quad \Longleftrightarrow \quad \text{Cons. share} = \frac{\varepsilon_S}{\varepsilon_S - \varepsilon_D}$$

**Reading.** The more inelastic the consumer relative to supply ( $|\varepsilon_D|$  small vs  $\varepsilon_S$  large), the higher the consumer share. Incidence is entirely determined by the ratio of the two elasticities.

## Incidence: The Four Extreme Cases



- $\epsilon_D = 0$  or  $\epsilon_S \rightarrow \infty$ : consumers pay the whole tax (quantity does not change or supply is infinitely elastic).
- $\epsilon_S = 0$  or  $\epsilon_D \rightarrow -\infty$ : producers pay the whole tax (supply is fixed or consumers have perfect alternatives).

**Application.** Excise duties on gasoline and tobacco (inelastic demand) fall almost entirely on consumers. Land taxation (completely inelastic supply) falls entirely on landowners—a classic argument in favor of land taxes (Henry George).

# Application: Excise Duties on Tobacco and Gasoline

## Excise duties on tobacco — empirical incidence

- Estimated demand elasticity:  $|\varepsilon_D| \approx 0.3\text{--}0.5$  (adults),  $\approx 1$  (teenagers, more price-sensitive)
- Supply elasticity: very high,  $\varepsilon_S \approx 5$  in the LR (production is easily expandable)
- **Consumer share:**  $\varepsilon_S/(\varepsilon_S - \varepsilon_D) = 5/(5 + 0.4) \approx 93\%$
- In Italy, the tobacco excise accounts for about 58% of the retail price, and Bank of Italy studies find almost complete pass-through to consumers

## Excise duties on gasoline — similar

- SR elasticity:  $|\varepsilon_D| \approx 0.1\text{--}0.3$ ; LR elasticity:  $|\varepsilon_D| \approx 0.5\text{--}0.8$
- Consumers cannot quickly change car or residence in response to the price
- Consumer share  $> 80\%$  in almost all estimates

## A debated normative implication

Taxing inelastic goods (tobacco, gasoline, alcohol, sugars) is *efficient*—minimum DWL per unit of revenue—but often *regressive*: inelastic goods weigh more in the basket of low-income households. The efficiency/equity trade-off reappears in acute form.

## The Tax Wedge on Labor in Italy (1/2)

A direct application of the tax wedge to the Italian labor market: the difference between the firm's labor cost and the worker's net pay.

**Typical breakdown for a gross salary of €30,000:**

|                                      |                              |
|--------------------------------------|------------------------------|
| Labor cost for the firm              | ≈ 40,000                     |
| of which employer INPS contributions | ≈ 10,000                     |
| Gross pay                            | 30,000                       |
| of which IRPEF + local surcharges    | ≈ 6,000                      |
| of which employee INPS contributions | ≈ 3,000                      |
| <b>Net pay to the worker</b>         | <b>21,000</b>                |
| <hr/>                                |                              |
| Total tax wedge                      | 19,000 (≈ 47% of labor cost) |

## The Tax Wedge on Labor in Italy (2/2)

**Microeconomic reading.** The wedge is a tax on labor, perfectly analogous to the model of this lecture:

- $P_d^*$  = labor cost for the firm = 40k (price paid by the “buyer of labor”)
- $P_s^*$  = net pay to the worker = 21k (price kept by the “seller of labor”)
- Total wedge = 19k, which the government splits between social contributions and IRPEF

**Actual incidence.** Labor supply is inelastic in the SR (hard to change jobs quickly), more elastic in the LR (migration, participation choices, education). In Italy the wedge is among the highest in the OECD (47% vs an average of 35%), with estimated effects of reducing labor force participation by 3–5 percentage points and a *tax wedge* that discourages the emergence of informal work.

## DWL and the Tax Rate: A Quadratic Relationship

With linear curves, DWL grows with the *square* of the tax rate.

**Intuition.** When  $t$  doubles:

- The reduction in quantity ( $Q^* - Q_t^*$ ) doubles:  $\Delta Q \propto t$
- The height of the DWL triangle doubles: it equals  $t$
- DWL is  $\frac{1}{2} \cdot \Delta Q \cdot t \propto t^2$

| $t$ | $Q_t^*$ | $Q^* - Q_t^*$ | DWL  | Ratio vs $t = 2.5$ |
|-----|---------|---------------|------|--------------------|
| 2.5 | 45      | 3             | 3.75 | —                  |
| 5   | 42      | 6             | 15   | $\times 4$         |
| 10  | 36      | 12            | 60   | $\times 16$        |
| 20  | 24      | 24            | 240  | $\times 64$        |

### Implication for fiscal policy

Doubling the tax rate **quadruples** the deadweight loss. It is more efficient to use many small taxes on different bases than one large tax on a single base. This principle (*broad base, low rate*) is one of the cornerstones of optimal taxation theory.

# What Should Be Taxed? Intuition from DWL

**For a given revenue  $\bar{T}$ , taxing which good minimizes DWL?**

For fixed  $\bar{T}$ , reducing the distorted quantity  $\Delta Q = Q^* - Q_t^*$  means increasing the price shares that separate  $P_d$  and  $P_s$ . For linear curves:

$$DWL \approx \frac{1}{2} \cdot \Delta Q \cdot t, \quad \bar{T} = t \cdot Q_t^* \approx t \cdot Q^*$$

Eliminating  $t = \bar{T}/Q^*$  and using  $\Delta Q \approx t \cdot \frac{Q^*}{P^*} \cdot \frac{\varepsilon_S |\varepsilon_D|}{\varepsilon_S + |\varepsilon_D|}$  (quantity falls more if demand and supply are elastic):

$$DWL \approx \frac{1}{2} \cdot \frac{\bar{T}^2}{Q^* \cdot P^*} \cdot \frac{\varepsilon_S |\varepsilon_D|}{\varepsilon_S + |\varepsilon_D|}$$

## Rule of thumb

To minimize DWL for a given revenue: tax goods with **inelastic** demand and supply. The distortion is directly linked to how much consumers and producers respond to the price—if they respond little, the quantity traded changes little, and the DWL triangle stays small.

**Equity limit.** This logic leads to taxing necessities (bread, energy, medicines)—the inelastic-demand goods par excellence. But they are also the goods that weigh most on low incomes: the efficient outcome can be strongly regressive. Real tax systems balance efficiency and equity.

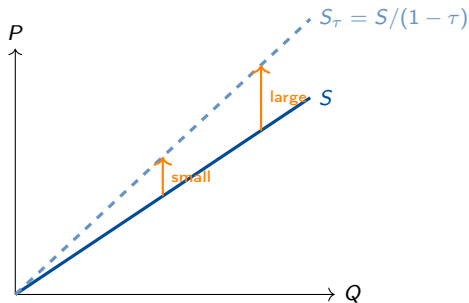
## Ad Valorem Tax: Mechanism (1/2)

### Ad valorem tax ( $\tau\%$ of the price)

The producer remits a share  $\tau$  of the gross price  $P_d$  to the government, keeping  $(1 - \tau)P_d$ . It agrees to produce if  $(1 - \tau)P_d \geq MC(Q)$ .

**Graphical difference from the specific tax.** The supply curve does not *shift* upward, it **rotates**: the vertical distance between the original  $S$  and  $S_\tau$  increases with the price.

## Ad Valorem Tax: Mechanism (2/2)



at low  $P$ , tax =  $\tau P$  small  
at high  $P$ , tax =  $\tau P$  large

**Practical consequence.** The ad valorem tax adjusts automatically to inflation: if prices rise by 10%, nominal revenue grows proportionally without having to revise the law. The specific tax must be updated periodically (see fuel excise duties, which the Italian legislature updates through the budget law).

# Ad Valorem Tax: Local Equivalence

## Point-by-point equivalence

For every specific tax  $t$  that yields an equilibrium with price  $P_d^*$ , there exists an ad valorem rate  $\tau^* = t/P_d^*$  that generates the **same** final equilibrium. All areas (CS, PS,  $T$ , DWL) are identical.

**Check in our example.** With  $t = 5$  we get  $P_d^* = 19$ ,  $P_s^* = 14$ ,  $Q_t^* = 42$ ,  $T = 210$ .

The equivalent ad valorem rate is  $\tau^* = 5/19 \approx 26.3\%$ . At that rate, the producer keeps  $(1 - \tau^*)P_d^* = 0.737 \cdot 19 \approx 14 = P_s^*$  ✓.

## Italian VAT at 22%

VAT in Italy is an ad valorem tax on the consumer (not on the producer): on a net price  $P_s$  of 100, the consumer pays  $P_d = P_s(1 + \tau) = 122$ . With  $\tau = 0.22$ , the share  $\tau$  goes to the government.

For products with inelastic demand (service subscriptions, energy): almost complete pass-through, the consumer pays the entire VAT. For products with elastic demand (non-essential luxury goods): the producer can absorb part of the increase.

**Why prefer ad valorem?** Besides automatic indexation: if the market is heterogeneous (goods with different prices), a uniform rate  $\tau$  yields revenue that increases with the price—fairer than the specific tax (the same for a good costing 10 and one costing 100).

## Tariffs and Quotas: Two Tools to Limit Imports (1/2)

Governments typically restrict imports in two ways:

- **Tariff**: a specific tax  $t$  or ad valorem tax  $\tau$  on imported goods. A tool we have already studied.
- **Import quota**: a fixed cap  $\bar{Q}$  on the quantity that can enter the country. Whoever obtains the import license captures a *rent*.

**In our market with  $Q^D = 80 - 2P$ ,  $Q^S = 3P$  (domestic producers), quota  $\bar{Q} = 36$ :**

Domestic price:  $80 - 2P = 36 \Rightarrow P_{\text{quota}} = 22$ . Price at which domestic producers would sell the 36<sup>th</sup> unit:  $P_S(36) = 12$ . The wedge of 10 per unit is the **quota rent** that goes to the license holder.

## Tariffs and Quotas: Two Tools to Limit Imports (2/2)

|                         | Equilibrium | Tariff $t = 10$        | Quota $\bar{Q} = 36$ |
|-------------------------|-------------|------------------------|----------------------|
| Price paid by consumers | 16          | 22                     | 22                   |
| Quantity traded         | 48          | 36                     | 36                   |
| CS                      | 576         | 324                    | 324                  |
| PS (domestic producers) | 384         | 216                    | 216                  |
| Revenue / Rent          | 0           | $T = 360$ (government) | $R = 360$ (private)  |
| DWL                     | 0           | 60                     | 60                   |

### Formal equivalence, substantive difference

A tariff  $t = 10$  and a quota  $\bar{Q} = 36$  produce the *same* price, the same quantity, the same CS, PS, and DWL. The only difference is **who captures the rectangle of 360**: the government (revenue) with the tariff, the license holders (private parties) with the quota.

# Quotas vs Tariffs: Why Governments Often Choose Quotas

If the tariff is more transparent, why do we often observe quotas in practice?

## Tariff (tax):

- Revenue to the government
- Imports grow if domestic demand grows (elastic filter)
- Transparent in welfare analysis
- Preferred by economists
- Regulated by the WTO/GATT

## Quota $\bar{Q}$ :

- Rent to license holders
- Blocks imports even if demand grows (rigid filter)
- Opaque, politically malleable
- Preferred by domestic producers
- Harder to discipline

## Historical case: Japanese cars in the US, 1981

To avoid formal tariffs (prohibited by the GATT), Japan “voluntarily” agreed to limit exports to the US to 1.68 million units (*Voluntary Export Restraints*). Result: *Japanese* producers captured the quota rent, selling more expensive cars on the American market. The US obtained the opposite of the policy goal: less government revenue and higher prices for consumers.

# The Subsidy: Mechanism and Equilibrium

## Specific subsidy $s$

A **subsidy**  $s$  is a government transfer for each unit produced. The producer receives  $P_s = P_d + s$ : the price paid by the buyer plus the subsidy paid by the government.

**Effect on the supply curve:** it shifts **downward** by  $s$ . For each quantity, the producer accepts a price lower by  $s$ , because the government makes up the difference.

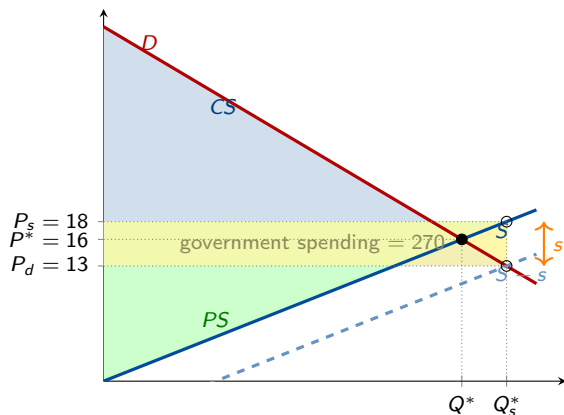
With  $s = 5$  in our market:

$$\begin{aligned}Q_s^S &= 3(P_d + 5) = 3P_d + 15 \\80 - 2P_d &= 3P_d + 15 \iff 65 = 5P_d \iff P_d^* = 13 \\P_s^* &= 13 + 5 = 18, \quad Q_s^* = 80 - 26 = 54\end{aligned}$$

**Compared with the equilibrium ( $P^* = 16$ ,  $Q^* = 48$ ):**

- Buyers pay  $-3$  per unit; producers receive  $+2$  per unit
- Quantity rises from 48 to 54
- Cost of the subsidy to the government:  $s \cdot Q_s^* = 5 \times 54 = 270$

## Subsidy $s = 5$ : Surplus, Fiscal Cost, and Deadweight Loss



### Equilibrium without subsidy

$$Q^* = 48, \quad P^* = 16$$

$$TS^* = \frac{1}{2} \cdot 48 \cdot 40 = 960$$

### Equilibrium with subsidy $s = 5$

$$Q_s^* = 54, \quad P_d = 13, \quad P_s = 18$$

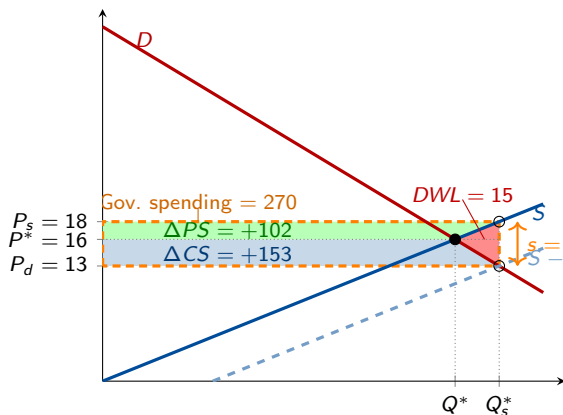
$$CS = \frac{1}{2} \cdot 54 \cdot (40 - 13) = 729$$

$$PS = \frac{1}{2} \cdot 54 \cdot 18 = 486$$

### Interpretation

The subsidy increases quantity beyond the efficient level. The gross gains of consumers and producers do not fully offset the fiscal cost.

## Subsidy $s = 5$ : Decomposing the Public Cost



The government pays a rectangle

$$\text{Cost} = s \cdot Q_s^* = 5 \cdot 54 = 270$$

Where do these 270 go?

- Consumers:  $\Delta CS = +153$
- Producers:  $\Delta PS = +102$
- Nobody:  $DWL = 15$

$$\underbrace{270}_{\text{Gov.}} = \underbrace{153}_{\Delta CS} + \underbrace{102}_{\Delta PS} + \underbrace{15}_{DWL}$$

### The point

The rectangle *leaves* the government's coffers; the blue and green areas *enter* private surpluses. But  $153 + 102 = 255 < 270$ : the red triangle is the public spending that becomes nobody's surplus.

## Subsidy: Why It Creates DWL (1/2)

The basic intuition is counterintuitive: a free transfer to producers increases their surplus *and* that of consumers (lower price, larger quantity). Yet it creates DWL. Why?

**Decomposition.** In our example with  $s = 5$ ,  $Q_s^* = 54$ :

- Consumers gain  $\Delta CS = +153$  (they buy more, at a lower price)
- Producers gain  $\Delta PS = +102$  (they sell more, at a higher price)
- Sum of private gains = 255
- The government spends  $\text{Cost} = 270$
- Private gain – Public cost =  $255 - 270 = -15 = \text{DWL}$

## Subsidy: Why It Creates DWL (2/2)

**Focus on units 49–54** (the “extra” ones the subsidy caused to be produced):

- Production cost: along curve  $S$  beyond  $Q^*$ , hence  $P_S(Q) > 16$
- Value to consumers: along curve  $D$  beyond  $Q^*$ , hence  $P_D(Q) < 16$
- Each unit has  $P_S(Q) > P_D(Q)$ : cost exceeds value
- The exchange generates a net *loss* for society

### Symmetry with the tax

The tax **prevents** socially useful trades (units between  $Q_t^*$  and  $Q^*$ :  $P_D > P_S$ ). The subsidy **forces** socially useless trades (units between  $Q^*$  and  $Q_s^*$ :  $P_D < P_S$ ). Both push  $Q$  away from  $Q^*$ , creating DWL of the same triangular shape.

# Subsidies in Practice: The CAP

## CAP direct payments (50 billion/year EU)

With the 1992 and 2003 reforms, the CAP replaced guaranteed prices with **direct payments per hectare**. Since 2003 they are subsidies *decoupled* from production: the farmer receives a fixed amount per hectare regardless of what he produces.

- Equivalent to a **lump-sum** per unit of land: less DWL than classic guaranteed prices
- The fiscal cost remains: 30% of the EU budget
- Distributional effects: a large share of payments goes to large landowners

# Subsidies in Practice: Electric Cars

## Ecobonus for electric cars (Italy 2020–2024)

Specific subsidy of up to €10,000 for the purchase of an electric car within price limits.

- Rationale: positive externality (emission reduction) and correction of coordination failures (charging network)
- **Incidence:** partial pass-through to consumers; producers raised prices, absorbing part of the subsidy (also documented with the US *Cash for Clunkers*)
- Distortions: it rewards purchases that would have happened anyway (*inframarginal*) + brings forward purchases from future years (*timing distortion*)

**General lesson.** A subsidy motivated by an externality should be sized as  $s^* = \text{marginal external benefit}$ . Subsidies more generous than that reproduce the DWL of our baseline example.

## Tax vs Subsidy: Full Comparison (1/2)

|                         | Equilibrium | Tax $t = 5$ | Subsidy $s = 5$ |
|-------------------------|-------------|-------------|-----------------|
| $P_d^*$                 | 16          | 19 (+3)     | 13 (-3)         |
| $P_s^*$                 | 16          | 14 (-2)     | 18 (+2)         |
| $Q^*$                   | 48          | 42 (-6)     | 54 (+6)         |
| CS                      | 576         | 441 (-135)  | 729 (+153)      |
| PS                      | 384         | 294 (-90)   | 486 (+102)      |
| Government revenue/cost | 0           | +210        | -270            |
| Net TS                  | 960         | 945         | 945             |
| DWL                     | 0           | <b>15</b>   | <b>15</b>       |

## Tax vs Subsidy: Full Comparison (2/2)

### Fundamental symmetry

A tax and a subsidy of the same size produce the **same** DWL: both move quantity by  $\pm 6$  relative to  $Q^*$ , with the same “base” and “height” of the DWL triangle.

**Why does the subsidy create DWL?** It encourages the production of units whose cost ( $MC$ ) exceeds their value to the consumer ( $P_D$ ). Those “extra” units cost society more than they are worth. The distortion has the *opposite* sign to the tax, but it is a distortion nonetheless.

## Exercise 1 — Specific Tax

Market with  $Q^D = 80 - 2P$ ,  $Q^S = 3P$ . Equilibrium:  $P^* = 16$ ,  $Q^* = 48$ ,  $TS = 960$ .  
The government imposes a specific tax  $t = 10$  on the producer.

- (a) Find  $P_d^*$ ,  $P_s^*$ ,  $Q_t^*$ .
- (b) Compute CS, PS, revenue  $T$ , and DWL.
- (c) Compute the tax incidence (consumer and producer shares).
- (d) Compare the DWL with that at  $t = 5$  ( $DWL = 15$ ). Is it consistent with  $DWL \propto t^2$ ?

## Exercise 1 — Solution (a) and (b)

(a)  $Q_t^S = 3(P_d - 10) = 3P_d - 30$ . Equilibrium:

$$80 - 2P_d = 3P_d - 30 \iff 110 = 5P_d \iff P_d^* = 22$$

$$P_s^* = 22 - 10 = 12 \quad Q_t^* = 80 - 44 = 36$$

(b)

$$CS = \frac{1}{2} \cdot 36 \cdot (40 - 22) = \frac{1}{2} \cdot 36 \cdot 18 = 324$$

$$PS = \frac{1}{2} \cdot 36 \cdot 12 = 216$$

$$T = 10 \times 36 = 360$$

$$DWL = \frac{1}{2}(48 - 36) \cdot 10 = 60$$

Overall check:  $CS + PS + T + DWL = 324 + 216 + 360 + 60 = 960 \checkmark$

## Exercise 1 — Solution (c) and (d)

**(c)** Incidence:

$$\text{Cons. share} = \frac{\Delta P_d}{t} = \frac{22 - 16}{10} = \frac{6}{10} = 60\%$$

$$\text{Prod. share} = \frac{16 - 12}{10} = \frac{4}{10} = 40\%$$

Same 60/40 incidence as the tax  $t = 5$  ✓. Incidence depends on the elasticities *at equilibrium*, not on the tax rate.

**(d)** Check of the quadratic relationship:

$$\frac{\text{DWL}(t = 10)}{\text{DWL}(t = 5)} = \frac{60}{15} = 4 = \left(\frac{10}{5}\right)^2 \quad \checkmark$$

Doubling the tax rate from 5 to 10 quadruples the deadweight loss from 15 to 60. Fully consistent with  $\text{DWL} \propto t^2$  (with linear curves).

## Exercise 2 — Equivalence Principle

Same market, same tax  $t = 5$ , but **on the buyer** (instead of the seller).

- (a) Write the new demand curve taking into account the tax on the buyer.
- (b) Find  $P_d^*$ ,  $P_s^*$ ,  $Q_t^*$ .
- (c) Compare with the narrative example (tax on the seller).

## Exercise 2 — Solution

(a) Tax on the buyer: for each unit, the buyer pays  $P_s$  to the producer and  $t$  to the government. The relationship becomes  $P_d = P_s + 5$ , where  $P_d$  is the total cost to the buyer. Demand:

$$Q_t^D = 80 - 2(P_s + 5) = 70 - 2P_s$$

(b)  $70 - 2P_s = 3P_s \implies P_s^* = 14, P_d^* = 19, Q_t^* = 42$ .

(c) **Identical** to the case of the tax on the seller:  $P_d^* = 19, P_s^* = 14, Q_t^* = 42, CS = 441, PS = 294, T = 210, DWL = 15$ .

### Equivalence Principle (strengthened)

The incidence of a tax does not depend on who formally remits it to the government, but only on the elasticities of demand and supply. The legal formulation of the tax is economically irrelevant.

## Exercise 3 — Subsidy

Same market. The government introduces a subsidy  $s = 10$  to the producer.

- (a) Find  $P_d^*$ ,  $P_s^*$ ,  $Q_s^*$ .
- (b) Compute CS, PS, cost to the government, and DWL.
- (c) Does the subsidy increase net welfare? Why does it create DWL?

## Exercise 3 — Solution

(a)  $Q_s^S = 3(P_d + 10) = 3P_d + 30$ .  $80 - 2P_d = 3P_d + 30 \Rightarrow 50 = 5P_d \Rightarrow P_d^* = 10$ .  $P_s^* = 20$ ,  $Q_s^* = 60$ .

(b)

$$CS = \frac{1}{2} \cdot 60 \cdot (40 - 10) = 900$$

$$PS = \frac{1}{2} \cdot 60 \cdot 20 = 600$$

$$\text{Cost} = 10 \cdot 60 = 600$$

$$TS_{\text{net}} = 900 + 600 - 600 = 900, \quad DWL = 960 - 900 = 60$$

Equivalently:  $\frac{1}{2}(60 - 48) \cdot 10 = 60$ .

(c) **No.** Net TS falls from 960 to 900. The subsidy pushes production beyond  $Q^*$ : the 12 additional units (from 48 to 60) cost producers more ( $P_s$  increasing along  $S$ ) than they are worth to consumers ( $P_D$  decreasing along  $D$ ). The government transfer (600) exceeds the private gain ( $\Delta CS + \Delta PS = 324 + 216 = 540$ ) by exactly 60.

## Exercise 4 — Incidence for Given Elasticities

In a generic market with  $\varepsilon_D = -0.4$  and  $\varepsilon_S = 0.6$  at equilibrium, the government introduces a small tax.

- (a) Using the incidence formula, determine the consumer and producer shares.
- (b) If we want consumers to bear at most 50% of the tax, which of the two elasticities must change, and in which direction?
- (c) Why do excise duties on tobacco fall almost entirely on consumers?

## Exercise 4 — Solution

(a)

$$\text{Cons. share} = \frac{\varepsilon_S}{\varepsilon_S - \varepsilon_D} = \frac{0.6}{0.6 - (-0.4)} = \frac{0.6}{1} = 60\%$$

$$\text{Prod. share} = 40\%$$

(b) We would need  $|\varepsilon_D| \geq \varepsilon_S$ , i.e., demand would have to become *more* elastic (in absolute value) than supply. Alternatively, a more inelastic supply.

In practice: one could promote substitutes for the good (raises  $|\varepsilon_D|$ ) or regulate producers to make them less flexible (lowers  $\varepsilon_S$ ). Both routes shift the incidence share.

(c) The demand for tobacco is strongly **inelastic** ( $|\varepsilon_D| \approx 0.4$ ) due to physiological dependence and the lack of close substitutes. Supply is much more elastic ( $\varepsilon_S$  large): the industry can reduce or expand production relatively easily. Consumer share:

$$\frac{\varepsilon_S}{\varepsilon_S + |\varepsilon_D|} \approx \frac{5}{5.4} \approx 93\%$$

## Exercise 5 — Comparing a Tax and a Subsidy

Same market  $Q^D = 80 - 2P$ ,  $Q^S = 3P$ . We want to compare two scenarios:

- A) Specific tax  $t = 5$  on the producer
  - B) Specific subsidy  $s = 5$  to the producer
- 
- (a) Compute  $\Delta P_d$  and  $\Delta P_s$  relative to  $P^* = 16$  in both cases.
  - (b) Verify that the DWL is identical in the two scenarios.
  - (c) Compute  $\Delta CS + \Delta PS$  (change in private surpluses) in both cases. Why do they differ even though the DWL is the same?
  - (d) If the government faces a budget constraint, which scenario costs more?

## Exercise 5 — Solution

(a) Tax:  $\Delta P_d = +3$ ,  $\Delta P_s = -2$ . Subsidy:  $\Delta P_d = -3$ ,  $\Delta P_s = +2$ . **Mirror-image movements.**

(b) Tax:  $\tilde{Q} = 42$ ,  $DWL = \frac{1}{2} \cdot 6 \cdot 5 = 15$ . Subsidy:  $\tilde{Q} = 54$ ,  $DWL = \frac{1}{2} \cdot 6 \cdot 5 = 15$ . ✓

Base and height of the triangle are equal: same distance from  $Q^*$  (6 units), same wedge (5).

(c)

- Tax:  $\Delta CS + \Delta PS = -135 - 90 = -225$  (private parties lose 225, the government gains 210, DWL 15)
- Subsidy:  $\Delta CS + \Delta PS = +153 + 102 = +255$  (private parties gain 255, the government spends 270, DWL 15)

The movements in absolute value differ because the wedge rectangle (of side  $t$  or  $s$ ) is subtracted from private surpluses (tax) or added to them (subsidy). But the *net balance* (private + government) is identical:  $-15$ , i.e., the DWL.

(d) The subsidy costs the government 270, while the tax yields 210. One is an outlay, the other a receipt. The subsidy has a higher *opportunity cost*: it must be financed with other taxes (which in turn have DWL). This is a general lesson: subsidies are intrinsically more costly than taxes of equal size.

# General Summary

## Types of tax

- Lump-sum: no distortion ( $DWL = 0$ ), but often unfair
- Specific ( $t$  per unit): curve  $S$  shifts vertically upward by  $t$
- Ad valorem ( $\tau\%$ ): curve  $S$  rotates upward; local equivalence  $\tau^* = t/P_d^*$

## Specific tax $t$

- $P_d^* > P^* > P_s^*$ , with  $P_d^* - P_s^* = t$ ;  $Q_t^* < Q^*$
- $T = t \cdot Q_t^*$ ;  $DWL = \frac{1}{2}(Q^* - Q_t^*)t \propto t^2$

## Tax incidence

- Does not depend on who remits the tax (Equivalence Principle)
- The less elastic side bears the larger share
- Formula: Cons. share  $= \varepsilon_S / (\varepsilon_S - \varepsilon_D)$

## Subsidy

- Mirror-image effect of the tax:  $Q_s^* > Q^*$
- DWL identical to a tax of the same size
- Extra units (between  $Q^*$  and  $Q_s^*$ ) cost more than they are worth

## Skills Acquired

- ✓ Distinguish specific, ad valorem, and lump-sum taxes
- ✓ Compute  $P_d^*$ ,  $P_s^*$ ,  $Q_t^*$  after a specific tax
- ✓ Compute CS, PS, revenue, and DWL in the presence of a tax
- ✓ Determine tax incidence with the shares  $\Delta P_d/t$  and  $|\Delta P_s|/t$
- ✓ Relate incidence to elasticities ( $\varepsilon_S/(\varepsilon_S - \varepsilon_D)$ )
- ✓ Apply the Equivalence Principle (irrelevance of who remits)
- ✓ Recognize the quadratic relationship  $DWL \propto t^2$
- ✓ Analyze a subsidy as the mirror image of a tax and compute its DWL

## Optional Homework Exercises

1.  $Q^D = 100 - 2P$ ,  $Q^S = 3P$ . Compute the equilibrium, CS, PS, TS. With a tax  $t = 5$  on the producer:  $P_d^*$ ,  $P_s^*$ ,  $Q_t^*$ ,  $T$ , DWL.
2. In the market of item 1, compute  $\varepsilon_D$  and  $\varepsilon_S$  at equilibrium and verify the incidence formula.
3. In a market with  $|\varepsilon_D| = 0.3$  and  $\varepsilon_S = 0.7$ , who bears the larger share of a tax? Compute the shares.
4. Explain why a lump-sum tax is efficient but politically unpopular. Recall the example of the British Poll Tax.
5.  $Q^D = 120 - 3P$ ,  $Q^S = 5P$ . The government introduces a subsidy  $s = 4$ . Compute  $P_d^*$ ,  $P_s^*$ ,  $Q_s^*$ , the cost to the government, and DWL.
6. The government wants to raise revenue of 200 with minimum DWL. It has two markets, A and B, both with perfectly elastic supply, but  $|\varepsilon_D^A| = 0.5$  and  $|\varepsilon_D^B| = 2$ . Which one should it tax? Justify.